MAPS FROM A PERFECT OBJECT INTO AN OBJECT SUPPORTED ON V(F) ARE ANNIHILATED BY A POWER OF F

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Mayer-Vietoris for Hom in the derived category and the induction principle for quasi-compact opens. invoked at proof lines 3657-3662. Reduces the assertion to the affine case by gluing over unions.
- On an affine scheme the derived category of quasi-coherent modules is equivalent to the derived category of modules, and perfect objects correspond to bounded complexes of finite projective modules. invoked at proof lines 3665-3670. Provides the module representation of E and the translation of alpha.
- The stupid truncation distinguished triangle of a bounded complex exhibiting the last term. invoked at proof lines 3672-3675. Supports the induction on the length of P^bullet reducing to a single term.
- Elements of a module supported on V(f) are annihilated by a power of f, and a map from a finite module into a module whose elements are f-power torsion is annihilated by a single power of f. invoked at proof lines 3682-3685. Concludes the annihilation in the single-term case.

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 10581–10581. Cited at proof lines 3657-3658 to glue the assertion over unions of opens; the reduction to the affine case is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Cited at proof lines 3661-3662 to reduce to X affine; the affine case assembly is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 3666-3667 to translate the statement into modules; the induction and annihilation are inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 3669-3670 to represent E by P^bullet; the length induction and annihilation are inside the counted span.

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Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $f \in \Gamma(X, \mathcal{O}_X)$. For any morphism $\alpha : E \to E'$ in $D_{QCoh}(\mathcal{O}_X)$ such that*

(1) *E is perfect, and*
(2) *E' is supported on $T = V(f)$*

there exists an $n \geq 0$ such that $f^n\alpha = 0$.

Proof. We have Mayer-Vietoris for morphisms in the derived category, see Cohomology, Lemma 08BW. Thus if $X = U \cup V$ and the result of the lemma holds for $f|_U$, $f|_V$, and $f|_{U\cap V}$, then the result holds for f. Thus it suffices to prove the lemma when X is affine, see Cohomology of Schemes, Lemma 08DR.

Let $X = \mathrm{Spec}(A)$. Then $f \in A$. We will use the equivalence $D(A) = D_{QCoh}(X)$ of Lemma 06Z0 without further mention. Represent E by a finite complex of finite projective A-modules $P^\bullet$. This is possible by Lemma 08EB. Let t be the largest integer such that P^t is nonzero. The distinguished triangle

$$P^t[-t] \to P^\bullet \to \sigma_{\leq t-1}P^\bullet \to P^t[-t+1]$$

shows that by induction on the length of the complex $P^\bullet$ we can reduce to the case where $P^\bullet$ has a single nonzero term. This and the shift functor reduces us to the case where $P^\bullet$ consists of a single finite projective A-module P in degree 0. Represent E' by a complex $M^\bullet$ of A-modules. Then α corresponds to a map $P \to H^0(M^\bullet)$. Since the module $H^0(M^\bullet)$ is supported on $V(f)$ by assumption (2) we see that every element of $H^0(M^\bullet)$ is annihilated by a power of f. Since P is a finite A-module the map $f^n\alpha : P \to H^0(M^\bullet)$ is zero for some n as desired. □